

JUAN DANIEL RAMIREZ

Senior Applied AI Engineer | RAG, AI Agents & LLM Evaluation | Healthcare, FinTech & EdTech
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PROFESSIONAL SUMMARY

Senior Applied AI Engineer with 7+ years building secure, regulated platforms across healthcare, FinTech, EdTech, and marketplaces, most recently shipping multi-agent RAG systems, production LLM features with guardrails and evals, and the data pipelines that feed them. Engineered regulated digital banking services supporting more than 500,000 users. Measurable impact includes a 30–40% cut in clinician documentation time and 120+ hours of manual work automated monthly.

TECHNICAL SKILLS

Languages & Frameworks: Python, TypeScript, JavaScript, Java, SQL, FastAPI, Node.js, Express.js, React

AI & LLMs: OpenAI, Claude, Claude Code, Gemini, Vertex AI, LangChain, LangGraph, Multi-Agent Systems, A2A, MCP

Retrieval & LLMops: Hybrid RAG, Embeddings, Vector Search, BM25, Knowledge Graphs, LLM Evals, Langfuse, Guardrails

Data & Messaging: PostgreSQL, pgvector, Redis, BigQuery, Feast, ETL/ELT Pipelines, Spring Batch, Celery, RabbitMQ

Cloud & DevOps: Google Cloud, AWS, Azure, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD

Security & Compliance: OAuth 2.0, RBAC, PII Redaction (Presidio), HIPAA, NIST AI RMF, Audit Logging

PROFESSIONAL EXPERIENCE

Lindy — San Francisco, California | Remote
Applied AI Engineer

Feb 2025 – Present

- Led end-to-end architecture and production deployment of a multi-agent hybrid RAG platform for clinical workflows, integrating EHR context, enterprise knowledge graphs and vector retrieval to cut clinician documentation time by 30–40% with HIPAA-compliant auditability.
- Implemented a hybrid retrieval layer combining BM25 keyword search, dense vector retrieval, structured metadata filtering and graph-linked longitudinal records, improving answer relevance and reducing hallucinations in clinical assistant responses.
- Designed and implemented agent orchestration in LangChain and LangGraph on the Agent2Agent (A2A) protocol with a lightweight Model Context Protocol (MCP) layer, standardizing state handoffs, tool authorization and encrypted context passing across voice-capture, speech-to-text, summarization and EHR-intent agents.
- Established an agent evaluation framework (A/B testing, model evals, behavioral drift detection, automated rollback triggers) that shortened incident detection and supported safe continuous deployment under AI governance policies.
- Built GCP-native inference pipelines (Vertex AI, Anthos) for hybrid LLM deployment, with on-prem/cloud routing and low-latency autoscaling for PHI-sensitive workloads under HIPAA and NIST AI RMF guardrails.
- Designed feature stores and offline evaluation pipelines (Feast, BigQuery) that keep tabular features for traditional ML models versioned and auditable, with HIPAA-compliant reproducibility across both the ML models and the LLM agents.
- Used Claude Code for AI-assisted development of agent services, speeding up test writing and refactoring while enforcing human code review and PHI-safe prompting rules for HIPAA compliance.
- Mentored cross-functional engineering teams on agentic design patterns and partnered with ITG and vendors to move ambient documentation features into secure, auditable production.

Acquire.com — San Francisco, California | Remote
AI Engineer

Jul 2024 – Feb 2025

- Shipped AI-generated, listing-specific questions to help buyers start conversations with sellers: FastAPI sends listing data to a lightweight OpenAI model that returns 3–5 questions as structured JSON output with Pydantic schema validation.
- Added LLM guardrails (retry and generic-question fallback), per-listing Redis caching invalidated only on seller edits to cut inference costs, Langfuse tracing of every prompt and output, and an A/B test on buyer–seller replies.
- Built an AI due diligence task manager: on offer acceptance, a Celery job runs PII redaction with Microsoft Presidio, then a stronger OpenAI model adapts an M&A advisor-written checklist to the business type (grounded generation).

- Validated each task (title, category, owner, due date) with Pydantic before saving to PostgreSQL, plus retry and checklist fallback so no deal lacks tasks; refined prompts via human-in-the-loop edit tracking and LLM evals on sample deals.
- Designed semantic buyer-listing matching on OpenAI embeddings in PostgreSQL + pgvector, pairing vector similarity search with budget and business-type filters and LLM match explanations; measured invites and offers against filters alone.
- Provisioned the infrastructure for all three AI features (FastAPI, Celery, Redis, PostgreSQL + pgvector) with Terraform.

Orbital Education — Dallas, Texas | On-site

Jun 2021 – Jun 2024

AI Engineer

- Built an LLM chatbot that lets staff ask plain-English questions about attendance, grades, and student records for 5,000+ students, answering from the PostgreSQL reporting layer via OpenAI function calling with role-based access control.
- Developed rubric-based LLM grading for short-answer and essay responses, returning scores and feedback as structured output that teachers review and approve before grades are recorded (human-in-the-loop).
- Engineered an AI document processing pipeline that extracts student data from enrollment forms and transcripts with OCR and LLM field extraction, validates it, and loads it into student records via Spring Batch, replacing manual data entry.
- Designed Spring Batch ETL pipelines on Quartz schedules, consolidating student information, attendance, and grading data into audit-ready PostgreSQL reporting models and processing roughly 30,000 records nightly.
- Hardened pipelines with restartable job steps, skip policies, and failure alerting, replacing manual spreadsheet exports and eliminating over 120 hours of reporting work each month.
- Led a team of three engineers and collaborated with QA across technical planning, mentoring, Java code reviews, quality processes, and production delivery.

Q2 Holdings — Austin, Texas | On-site

May 2020 – May 2021

Software Engineer

- Engineered TypeScript (Node.js) and Python microservices for a regulated digital banking platform serving more than 500,000 users across authentication, account management, investment, and transaction-processing workflows.
- Developed secure REST APIs on AWS using OAuth 2.0, PostgreSQL, and Redis to support financial operations across web, iOS, and Android applications.
- Implemented RabbitMQ-based asynchronous messaging to decouple transaction events and feed downstream processing across banking services.
- Diagnosed Celery and Redis bottlenecks with Flower and Redis SLOWLOG, then routed long-running tasks to dedicated queues, tuned worker concurrency and prefetch, and added pipelining and connection pooling to clear backlogs at peak load.
- Partnered with product, compliance, and operations teams to translate regulatory requirements into secure technical designs and production-ready banking features delivered quarterly.
- Helped implement CI/CD pipelines with Jenkins, GitHub Actions, and Docker and automated testing standards adopted by more than 15 internal teams, reducing deployment risk and improving release efficiency.

Bottle Rocket — Dallas, Texas | On-site

Mar 2019 – May 2020

Software Engineer, Part-Time

- Maintained and modernized client web apps using React and Node.js, improving stability, performance, and maintainability.
- Built and maintained REST APIs with Express.js on Node.js to power React front ends and integrate third-party services.
- Diagnosed and resolved performance bottlenecks across React and Node.js code, fixing memory leaks, blocking event-loop work, slow API responses, and unnecessary re-renders.
- Profiled CPU, memory, and rendering with Chrome DevTools, the React Profiler, and the Node.js inspector, then refactored slow components and endpoints.
- Partnered with client teams to prioritize production issues and ship reliable releases via GitHub Actions CI/CD to AWS.

EDUCATION

Bachelor's Degree in Computer Science

Aug 2016 – May 2020

Fontbonne University | St. Louis, Missouri